

# “Self-Healing Concrete: An Innovative Solution To Structural Cracking

Ajay Kumar Singh<sup>1</sup>, Ankul Sharma<sup>2</sup>, Ayush Singh<sup>3</sup>, Masoom Ahtesham<sup>4</sup>,  
Vijay Kumar Srivastava<sup>5</sup>

<sup>1,2,3,4</sup>B.Tech 4<sup>th</sup> year students, Department of Civil Engineering, Buddha Institute of Technology, Gorakhpur, UP, India

<sup>5</sup>Asst. Prof., Department of Civil Engineering, Buddha Institute of Technology, Gorakhpur, UP, India.

## ABSTRACT

Concrete is pretty tough and can handle high pressure, but it does need some steel reinforcement since it can't take on tensile forces that might weaken its overall strength. Plus, the surface cracks in reinforced concrete can be a real concern, as they pose risks to the entire structure. The development of self-healing concrete is especially important because it could reduce the need for expensive crack detection and repair costs in the long run. Self-healing concrete improves durability through its ability to prevent reinforcement corrosion and concrete deterioration which results in decreased maintenance expenses. The researchers conducted a pioneering study to investigate how microbiologically produced fillers which include the bacterium *Bacillus megaterium* affect the performance of concrete. The researchers created concrete cubes that contained bacterial inclusions and those which did not to evaluate how the bacterial presence affected their performance. The research demonstrated that *Bacillus megaterium* had exceptional concrete healing abilities which improved the strength of high-strength structural concrete. This study demonstrates that *Bacillus megaterium* functions as an effective agent which improves the strength and performance of concrete materials. The new method develops a sustainable solution which uses bacterial void reduction to strengthen concrete through improved mechanical attributes and enhanced durability. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives. The study shows *Bacillus megaterium* functions as an effective material which improves concrete strength and performance. The innovative method develops a sustainable solution which uses bacterial void reduction to improve the mechanical properties and complete durability of concrete. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives. The construction industry adopts new materials science developments which show potential to create self-healing concrete through the combination of microbiological additives.

**Keywords:** *Self-healing concrete, Crack remediation, Calcium carbonate production, Compressive strength enhancement, Microbial growth in concrete, Concrete durability, Sustainable construction materials, Biomineralization, Microbial activity in construction.*

## 1. INTRODUCTION

Bio-concrete functions as a self-repairing concrete which automatically fixes any cracks that develop. Dutch researcher and microbiologist Hendrik Jonkers developed the material using a special ingredient which produces automatic repairs without needing human repair work after installation. The construction industry will experience a complete transformation through the implementation of Bio-concrete technology. The construction process of self-healing concrete establishes bacterial contents, while the repair system and liquid system become operational only when the concrete elements sustain serious damages. Self-healing concrete represents the most advanced of the three concrete types. The cement mix includes bacterial spores which have been encapsulated in clay granules that measure between two and four millimeters and include nitrogen and phosphorus and a separate nutrient agent. This new method enables bacteria to remain inactive inside concrete for a duration of 200 years. The process takes place only when water enters a fissure because this method prevents contact with nutrients during the cement mixing procedure. The variant functions best for outdoor structures and for locations which present challenges for repair personnel. This method takes away the need for costly and complex manual maintenance tasks. Concrete is a popular construction material because its essential components are easily accessible. However, the main ingredient, cement, has an environmental impact, responsible for about 10 percent of the world's CO<sub>2</sub> emissions. Cracks in concrete can not only damage structures but also reduce the material's

strength and its ability to bear loads. This issue is worsened by extreme weather conditions, such as winter, which can cause water to seep in and freeze, widening cracks and further damaging the structure. The concrete needs immediate crack repairs because they protect small cracks from growing into larger cracks which would disrupt concrete functionality. Concrete is the most widely used man-made building material globally, created by mixing cementitious materials with aggregates or additives. Researchers currently investigate methods to decrease concrete usage through the implementation of waste material substitutes. Several studies have focused on reducing voids and permeability in concrete, and some have revealed the effectiveness of bacteria such as *Sporosarcina pasteurii* and *Bacillus licheniformis* in sealing cracks. Self-healing concrete (SHC) has become a revolutionary solution for addressing concrete cracking. The technique utilizes concrete's natural self-healing capacity because cracks will automatically repair themselves through continuous clinker mineral hydration or calcium hydroxide carbonation processes. The process can repair only small cracks because it needs water to operate. Researchers developed HAR with a bacterial stimulation system for crack sealing to address these restrictions. In 2006, self-healing concrete was invented by microbiologist Professor Henk Jonkers at Delft University of Technology in the Netherlands. The self-healing concrete (SHC) concrete system operates through its ability to automatically create healing material which functions as its self-healing mechanism. The concrete mixture includes designated materials which exist as either capsules or fibers that hold repair materials. The healing solution will be released through the rupture of capsules or fibers when cracks need to be sealed. The process of self-healing concrete (SHC) develops through two different methods which include (1) natural processes that occur with optimal cementitious material combinations and (2) specific healing agents which contain bacteria and additives. The presence of unhydrated cement particles in concrete enables small cracks of

0.2 mm size to heal through autogenous processes. Bacterial self-healing can repair cracks which have a maximum width of 0.8 millimeters. Bacterial self-healing concrete has shown through recent studies that it can repair cracks which measure between 0.3 millimeters and 1.5 millimeters. The bacterial agents which exist as embedded elements in concrete structures create a healing process that produces calcium carbonate precipitates which fill and seal all concrete cracks. The natural self-healing capacity of materials works effectively to repair small cracks which have widths that do not exceed 0.2 millimeters. The restoration process requires extra healing materials to fix larger cracks which breach structural integrity. The concrete industry needs to develop better sustainable production methods which will solve both cracking problems and environmental issues. Cement production serves as the main binding component of concrete which generates 4 to 7 percent of worldwide carbon dioxide emissions according to current data. The need for cement alternatives arises because companies need to cut their carbon dioxide emissions while creating environmentally friendly practices. China, which serves as the top seafood producer in the world, disposes

approximately 10 million tons of seashell waste from oysters clams scallops and mussels at landfills each year. The worldwide concrete demand reached 14 billion cubic meters in 2020, but it will grow to 20 billion cubic meters by 2050 according to World Cement and Concrete Association. The French Academy of Sciences published its first documentation of autogenous self-healing concrete technology in 1836. Early research established that concrete can naturally mend minor cracks through calcium carbonate deposition which occurs during the hydration process.

## 2. METHODOLOGY

### Preparation of Bacterial Concrete:

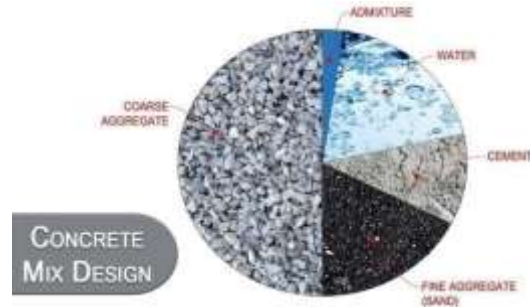
Direct application method: Bacterial spores and calcium lactate are added directly to the concrete during mixing. The concrete maintains its original properties because this process operates without changing its composition. The bacteria present in the system begin to germinate when water enters through the cracks which results in limestone production that closes the openings.



**Fig.1. *Bacillus megaterium* cells.**

### Material selection & preparation Materials:

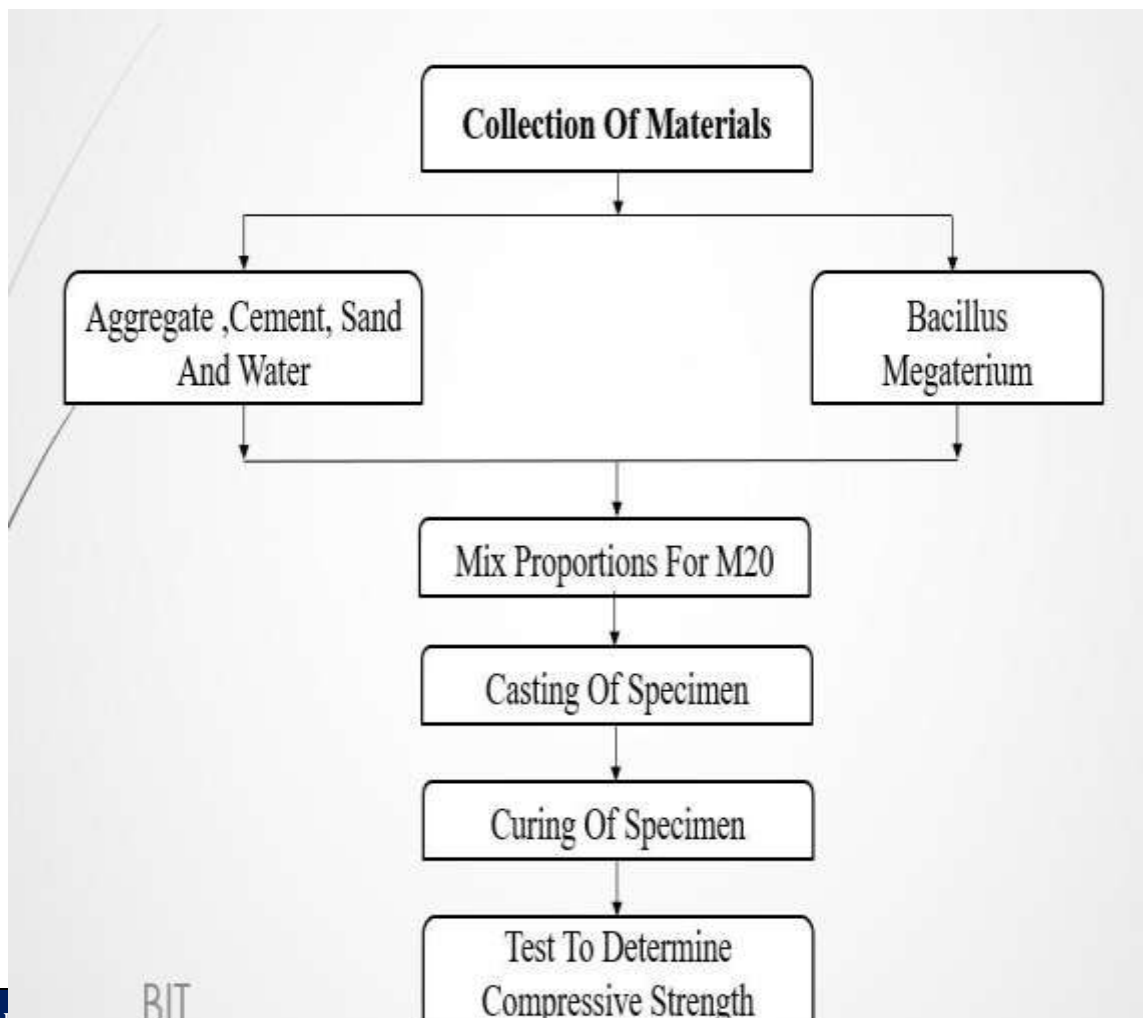
- Cement: Cement serves as the primary binding agent within concrete
- Coarse aggregate: Coarse aggregate constitutes the strongest and least porous component.
- Fine aggregate: Fine aggregate aids in water retention and minimizes void spaces.
- Water: Water is vital for hydration and acts as a binding agent.
- Bacteria: The specific type of bacteria utilized is *Bacillus megaterium*, which is incorporated in powdered form into the concrete mixture.
- Mix Design: Regarding mix design, it is essential to ascertain the appropriate proportions of cement, aggregate, and water to achieve the desired strength, with bacteria being added along with water during the mixing process.



**Fig.2. Concrete mix design**

### Mix Design Procedure:

- Material Preparation: It is essential to ensure that all materials are at room temperature. Thoroughly mix the cement and store it in a dry location. Additionally, allow the aggregate to dry and separate.
- Proportioning: Maintain proportions similar to those used on the construction site.
- Weighing: Accurately determine the quantities of cement, aggregate, and water by weight.
- Concrete mixing: Mix by hand or with a machine to avoid material loss.
- Compacting: Ensure full compaction without segregation or excessive bleeding.



**Fig.3. Methodology adopted.**

### Concrete Casting and Testing:

- Cast cubes of bacterial concrete with appropriate water-bacteria ratios.



**Fig.4. Preparation of bacterial concrete.**



**Fig.5. Casting of bacterial concrete cube.**

- Cure cubes at normal temperature, sun dry, and test compressive strength.
- Check healing property by drilling holes in cubes and observing for calcium carbonate formation.
- This process ensures proper preparation, mixing, and testing of bacterial concrete, assessing its strength and healing properties.

### 3. RESULT & CONCLUSION

**Table 1:** Compressive Strength of M20 Bacterial Concrete (at 7 days).

| Cube Number      | Weight (grams) | Load (kN) | Strength (MPa) |
|------------------|----------------|-----------|----------------|
| Cube Number: 1   | 8274           | 238       | 10.57          |
| Cube Number:2    | 8585           | 250       | 11.11          |
| Cube Number:3    | 8457           | 260       | 11.55          |
| Average Strength |                |           | 11.07          |

**Table 2:** Compressive Strength of M20 Bacterial Concrete (at 14 days).

| Cube Number    | Weight (grams) | Load (kN) | Strength (MPa) |
|----------------|----------------|-----------|----------------|
| Cube Number: 1 | 8384           | 328       | 14.57          |
| Cube Number:2  | 8658           | 340       | 15.11          |

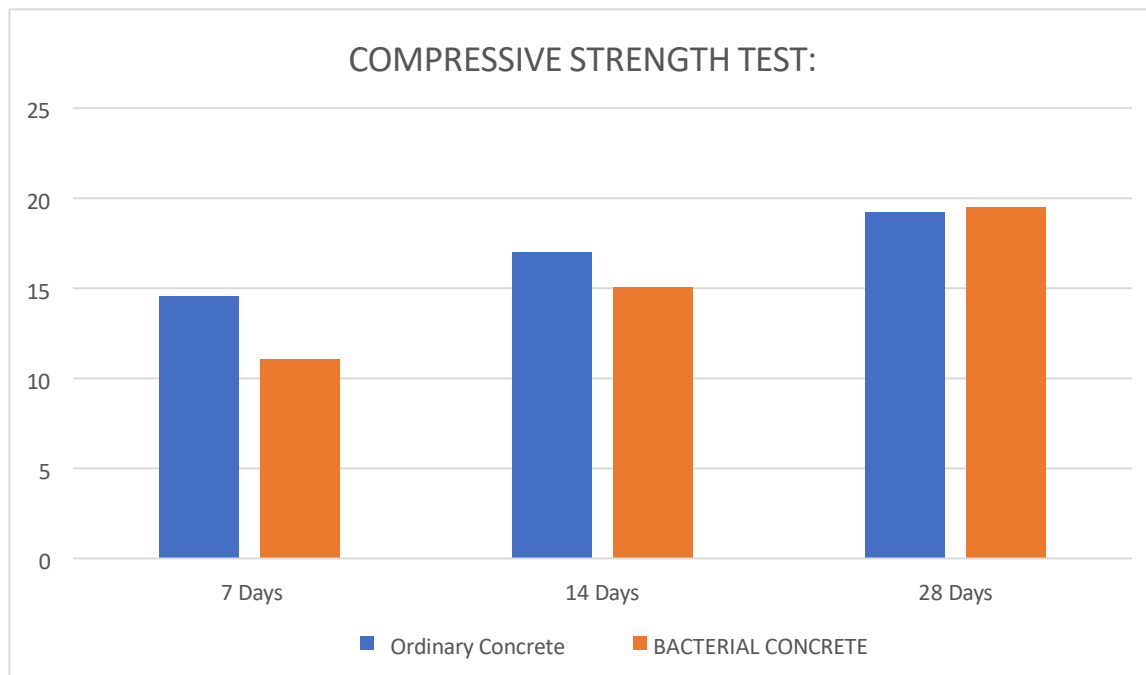
|                  |      |     |       |
|------------------|------|-----|-------|
| Cube Number:3    | 8427 | 350 | 15.55 |
| Average Strength |      |     | 15.07 |

**Table 3:** Compressive Strength of M20 Bacterial Concrete (at 28 days).

| Cube Number      | Weight (grams) | Load (kN) | Strength (MPa) |
|------------------|----------------|-----------|----------------|
| Cube Number: 1   | 8159           | 303       | 13.46          |
| Cube Number:2    | 8798           | 430       | 19.11          |
| Cube Number:3    | 8500           | 447       | 19.86          |
| Average Strength |                |           | 19.50          |

**Table 4:** Compressive Strength of M20 Ordinary Concrete.

| Days   | Weight (grams) | Load (kN) | Average Strength (MPa) |
|--------|----------------|-----------|------------------------|
| 7 Day  | 8649           | 328       | 14.57                  |
| 14 Day | 8585           | 382       | 16.97                  |
| 28 Day | 8680           | 432       | 19.20                  |



**Fig.6: Comparison of compressive strength of ordinary and bacterial M20**

## CONCLUSION

1. The article studies multiple methods which researchers use to create self-healing concrete through their research on bacterial application as a method to enhance the material's strength beyond standard concrete performance.
2. Various concrete-compatible bacteria that can be used for crack repair are addressed.
- 3 Bacteria closes gap in concrete via producing the essential mineral needed for sealing the cracks.
4. It has been discovered by researchers that bacteria enhance concrete cushioning properties.
5. The study examines different methods and techniques for designing self-healing concrete, classifying them into chemical, biological, and natural processes. The primary method for self-healing concrete development has relied on chemical techniques as its main approach.
6. The research investigates effective biological solutions which use bacteria as their primary research focus. The research proposes a complete system which categorizes self-healing concrete techniques to assist scientists in their studies.
7. The cost of repairing and maintaining concrete structures through traditional methods exceeds the expenses of using self-healing concrete. The existing methods require improvement because they need better execution to achieve concrete structure durability. The article presents design concepts and implementation techniques for self-healing concrete development.

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